

5 Factor VIII can be made as a recombinant human protein.

(a) (i) Name the disease that is treated using recombinant human factor VIII.

..... [1]

(ii) Before recombinant human factor VIII was available, this disease was treated with factor VIII from donated blood.

Give **two** advantages of using recombinant human factor VIII, instead of factor VIII from donated blood, to treat this disease.

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..... [2]

(b) The gene that codes for human factor VIII can be synthesised from messenger RNA purified from human liver cells.

(i) Name the enzyme that uses messenger RNA as a template to produce complementary DNA.

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(ii) Outline **two** sources, other than messenger RNA, from which genes can be obtained for genetic engineering.

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..... [2]

- (c) The gene that codes for human factor VIII can be transferred into mammalian cells in tissue culture.

Explain why a promoter also needs to be transferred into the mammalian cells so that human factor VIII can be synthesised.

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[Total: 8]